Prediction of Heart Disease Using Machine Learning Models

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Abstract

Heart disease is a major cause of mortality, motivating accurate and interpretable prediction systems. This project compares two supervised machine learning models—Random Forest and Support Vector Machine with the Kaggle *Heart Failure Prediction* dataset (918 observations, 12 variables).

Models were tuned via grid search with cross-validation and evaluated on a held-out test set using Accuracy, Sensitivity, Specificity, F1-score, and AUC. The SVM achieved the strongest test performance (Accuracy = 0.891, AUC = 0.944), while RF performed competitively. Model interpretability was assessed with model-agnostic XAI, including permutation feature importance, and feature effects.

Code and outputs: **GitHub Repository**.

Contents

1 Introduction

Cardiovascular disease remains a leading cause of mortality, making early and accurate risk prediction important for modern clinical decision support [6]. Machine learning methods can capture nonlinear relationships in demographic and clinical variables and may provide accurate classification from routinely collected measurements [1, 4, 5].

This study predicts the presence of heart disease and compares two widely used supervised classifiers: Random Forest [8] and Support Vector Machine with a radial basis function (RBF) kernel [9].

The analysis uses the Heart Failure Prediction dataset from Kaggle (Dataset Link), which contains 918 observations and 12 features with no missing values. The workflow includes exploratory analysis, data preprocessing, model training and tuning, evaluation on held-out test data, and model-agnostic interpretability analysis to identify influential predictors and explain model behaviour [11, 12].

2 Descriptive Analysis of the Data

The dataset contains 918 observations and 12 variables: 11 predictors (six numeric and five categorical) and the binary target *HeartDisease*. Class counts are 410 (no disease) and 508 (disease), indicating mild imbalance. No missing values are present. Categorical predictors were encoded as factors; numeric predictors were standardised using z-scores for comparable scaling.

2.1 Feature Description

Table 1: Feature Description Based on the Raw CSV Data

No.	**Feature**	**Data Type**	**Description**
1	Age	Numeric	Patient's age in years.
2	Sex	Nominal	Sex of the patient: `M` = Male, `F` = Female.
3	ChestPainType	Nominal	Chest pain type: `ATA` = Atypical angina, `NAP` = Non–anginal pain, `ASY` = Asymptomatic, `TA` = Typical angina.
4	RestingBP	Numeric	Resting blood pressure (mm/Hg).
5	Cholesterol	Numeric	Serum cholesterol level (mg/dl).
6	FastingBS	Nominal	Fasting blood sugar: 0 = $\leq$ 120 mg/dl, 1 = $>$ 120 mg/dl.
7	RestingECG	Nominal	Resting ECG results: `Normal`, `ST` = ST–T wave abnormality, `LVH` = Left ventricular hypertrophy.
8	MaxHR	Numeric	Maximum heart rate achieved.
9	ExerciseAngina	Nominal	Exercise-induced angina: `Y` = Yes, `N` = No.
10	Oldpeak	Numeric	ST depression induced by exercise relative to rest.
11	ST_Slope	Nominal	Slope of peak exercise ST segment: `Up` = Upsloping, `Flat` = Flat, `Down` = Downsloping.
Target Variable			
12	HeartDisease	Nominal	1 = Heart disease present, 0 = No heart disease.

2.2 Graphical Summary

Patients with heart disease tend to be older, reach lower MaxHR, and show higher Oldpeak. Across categorical predictors, ChestPainType, ExerciseAngina, and ST Slope show clear outcome separation (Figure 4), whereas RestingECG is weaker.

Overall, exercise-related variables (MaxHR, Oldpeak, ExerciseAngina, ST Slope) show the clearest separation and are expected to contribute strongly to prediction performance.

2.3 Statistical Hypothesis Testing

Welch's t-tests for numeric variables and chi-square tests for categorical variables were performed at a 5% significance level. Results are summarised in Table 2.

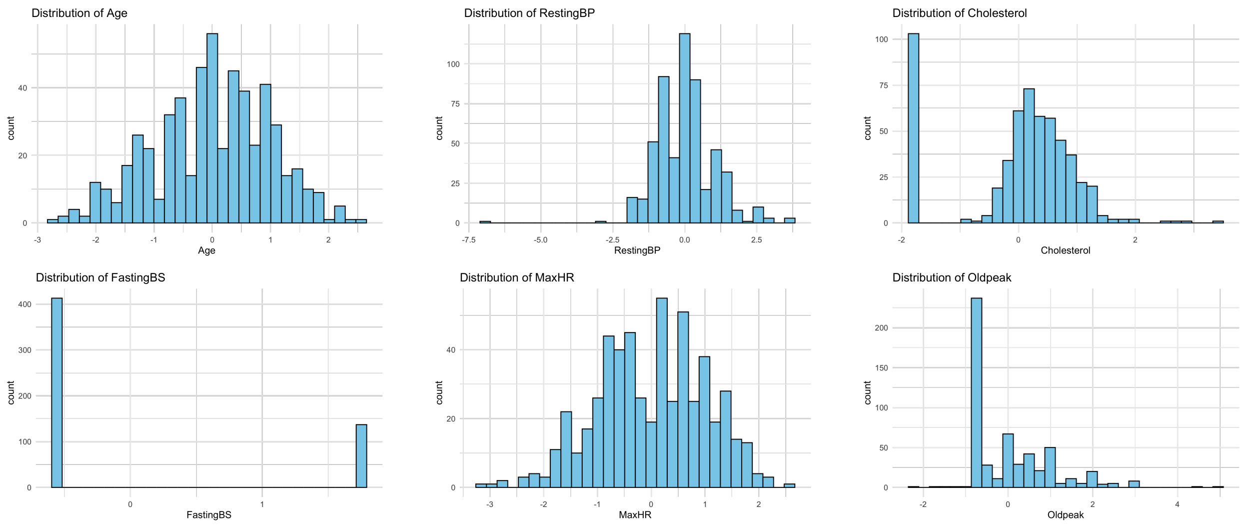


Figure 1: Numeric predictor distributions.

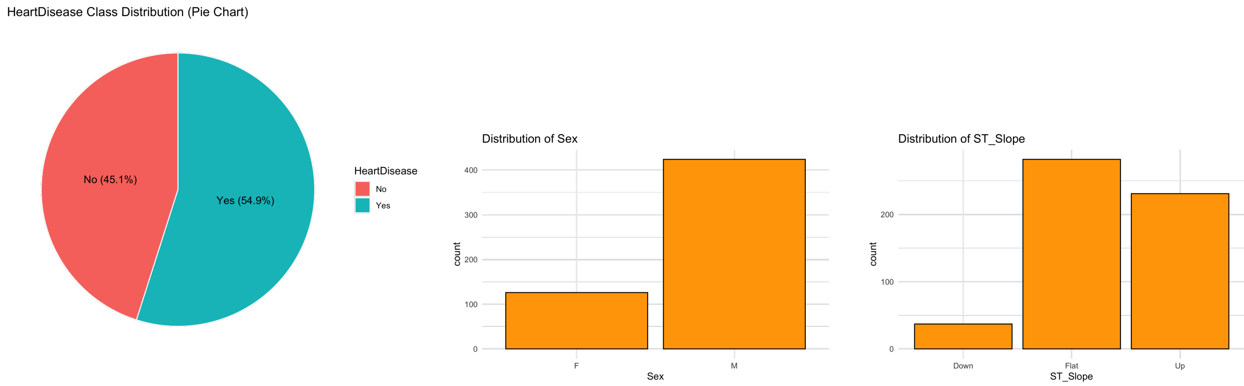


Figure 2: Class balance and key categorical distributions.

Figure 3: Key numeric predictors by outcome.

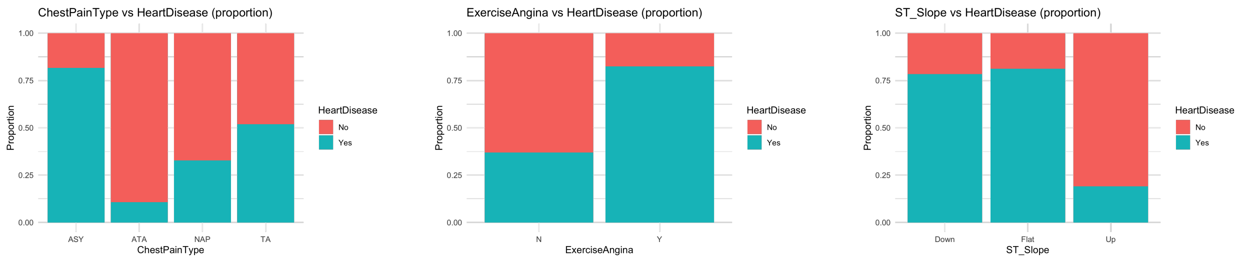


Figure 4: Outcome proportions across key categories.

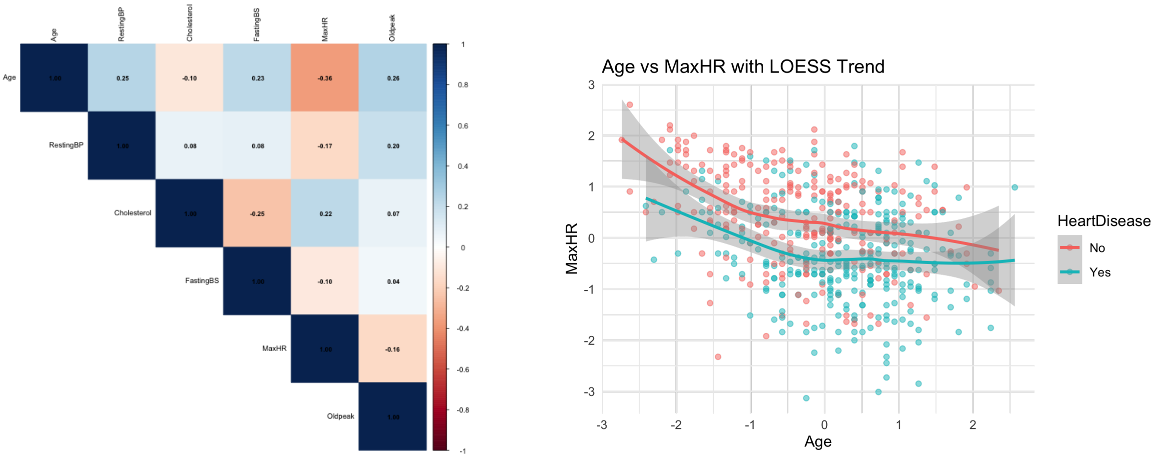


Figure 5: Correlation structure and Age–MaxHR trend.

Table 2: Statistical hypothesis testing of feature differences between patients with and without heart disease ($\alpha = 0.05$).

No.	Feature	Data Type	Statistical Test	p-value
1	Age	Numeric	t-test	$< 10^{-11}$
2	RestingBP	Numeric	t-test	0.002
3	Cholesterol	Numeric	t-test	$< 10^{-6}$
4	FastingBS	Numeric	t-test	$< 10^{-8}$
5	MaxHR	Numeric	t-test	≈ 0
6	Oldpeak	Numeric	t-test	≈ 0
7	Sex	Nominal	χ^2 test	$< 10^{-14}$
8	ChestPainType	Nominal	χ^2 test	≈ 0
9	ExerciseAngina	Nominal	χ^2 test	≈ 0
10	ST_Slope	Nominal	χ^2 test	≈ 0
11	RestingECG	Nominal	χ^2 test	0.051

3 Mathematical Overview of Machine Learning Methods

This section provides a brief mathematical overview of Random Forests and Support Vector Machines used in this study, focusing on their decision rules and key tuning parameters.

3.1 Random Forest

Random Forest [8] is an ensemble of decision trees trained on bootstrap samples, where each split considers only a random subset of features. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, $x_i \in \mathbb{R}^p$, $y_i \in \{0, 1\}$. Splits are chosen using the Gini impurity

$$G = 1 - (p_0^2 + p_1^2), \qquad p_k = \frac{N_k}{N}.$$

Each tree outputs a probability $T_b(x)$ for class 1, and the forest prediction is

$$\hat{p}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x), \qquad \hat{y}(x) = \mathbf{1}(\hat{p}(x) > \tau),$$

where τ is the decision threshold.

3.2 Support Vector Machine

Support Vector Machine (SVM) [9] is a supervised learning algorithm used for binary classification. It separates two classes by finding a linear decision boundary

$$f(x) = w^\top x + b,$$

where w is the weight vector and b is the bias term.

SVM aims to find a boundary with a large margin while allowing some classification errors. This is achieved by minimizing

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i,$$

where $\xi_i \geq 0$ are slack variables that allow misclassification and C controls the trade-off between margin size and classification errors.

For nonlinear data, SVM uses the radial basis function (RBF) kernel

$$K(x,z) = \exp(-\gamma\|x-z\|^2),$$

which maps the data into a higher-dimensional space.

The predicted label is given by

$$\hat{y}(x) = \mathbf{1}(f(x) \geq 0)\,.$$

4 Model Fitting, Tuning, and Comparison

All models were fitted using a consistent preprocessing pipeline and identical data splits to ensure a fair comparison. Hyperparameters were optimised via cross-validation, and performance was assessed on an independent test set using both threshold-independent and threshold-dependent metrics

4.1 Data Split and Preprocessing

The dataset was split into training (60%), validation (20%), and test (20%) subsets. Numeric predictors were standardised using training-set statistics (mean and standard deviation), and the same transformation was applied to the validation and test sets. Categorical predictors were treated as factors, with encoding handled internally by the modeling framework. Hyperparameters were tuned using 5-fold cross-validation on the training set, with ROC–AUC as the primary selection criterion. After tuning, the final model was retrained on the combined training and validation sets, and all final comparisons are reported on the held-out test set.

4.2 Hyperparameter Search and Final Model Choice

Table 3 summarises the hyperparameter grids considered for each method. The final configuration for each model was selected as the one achieving the highest mean cross-validated ROC–AUC on the training set.

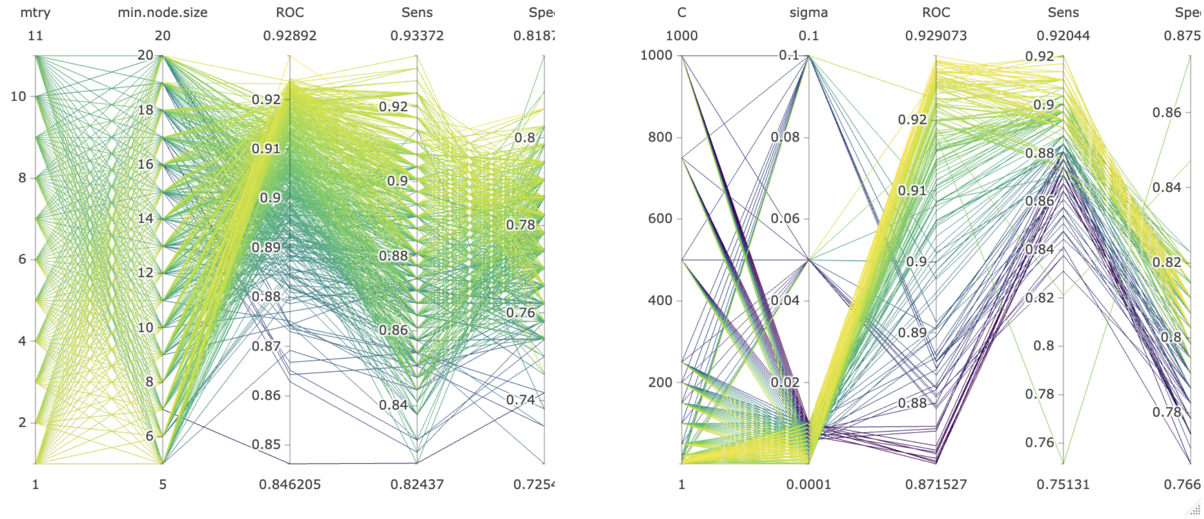
Model	**Hyperparameter**	**Values Explored**
Random Forest	mtry	{1,2,3,4,5,6,7,8,9,10}
	min.node.size	{3,5,7,9,11,13,15}
	splitrule	{`gini, extratrees, hellinger`}
SVM	Cost (C)	{1,5,10,20,50,100,150,200,250,500,750,1000}
	sigma (σ)	{0.1, 0.01, 0.05, 0.001, 0.002, 0.003, 0.004, 0.005, 0.006, 0.007, 0.008, 0.009, 0.0001}

Table 3: Hyperparameter grids used for tuning.

Figure 6 visualises the cross-validated ROC–AUC over the explored hyperparameter configurations for each model family.

4.3 Train–Validation Diagnostics

After selecting the final hyperparameter configuration for each model, we examined the agreement between training and validation performance as a diagnostic for potential overfitting. As shown in Figure 7, training and validation accuracies are close for both models, indicating a small generalisation gap. The SVM exhibits slightly equal validation accuracy, consistent with its stronger test-set performance.



(a) Random Forest tuning heatmap (b) SVM (RBF) tuning heatmap

Figure 6: Cross-validated tuning surfaces across hyperparameter settings.

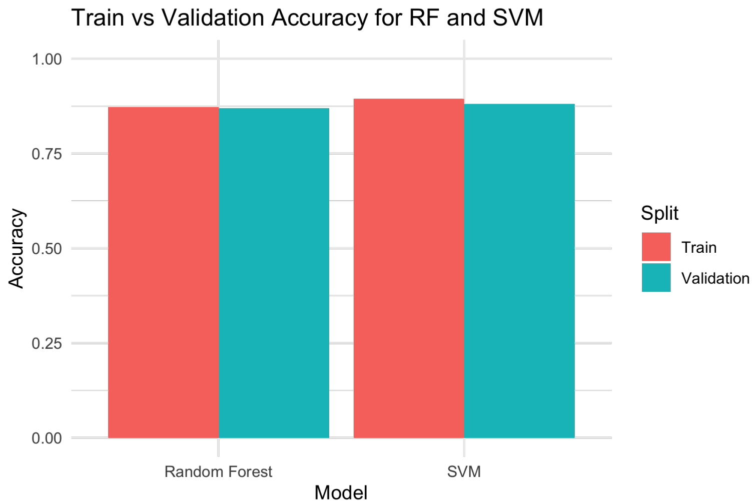


Figure 7: Training vs. validation accuracy for the selected configurations (diagnostic).

Decision threshold. For each model, the classification threshold was selected using Youden's index [10],

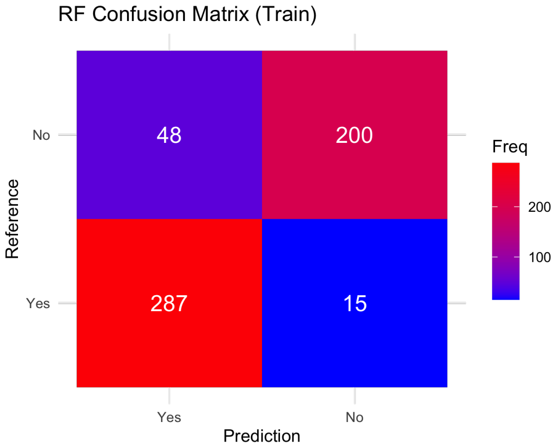
$$J = \text{Sensitivity} + \text{Specificity} - 1,$$

computed on pooled cross-validated (out-of-fold) training predictions. The resulting thresholds were $\tau = 0.4950$ for Random Forest and $\tau = 0.4750$ for SVM (RBF). These thresholds were used for all final class assignments reported below.

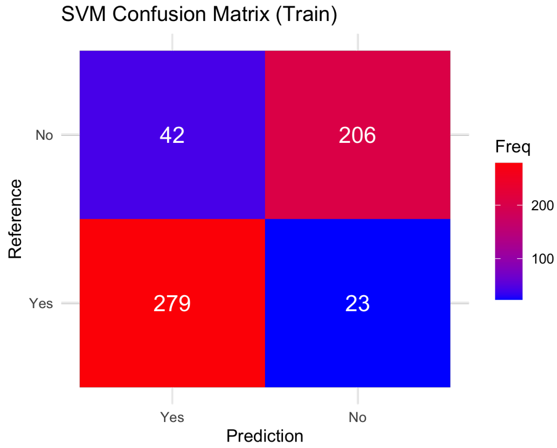
Evaluation metrics. We report Accuracy, Sensitivity (recall), Specificity, Precision, F1-score, and ROC–AUC. ROC–AUC summarises discrimination across all possible thresholds; higher values indicate better ranking of a randomly chosen positive instance above a randomly chosen negative instance.

4.4 Final Test Results

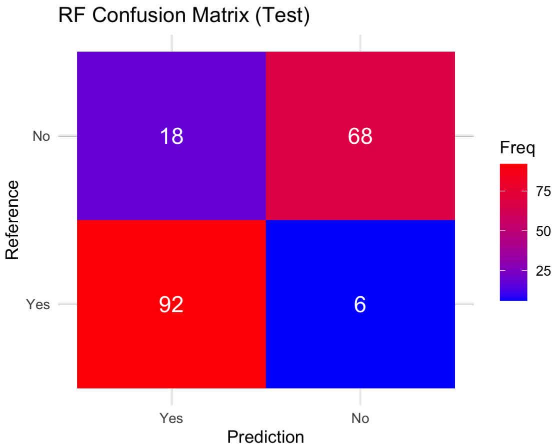
For SVM, decision scores were calibrated into class probabilities during training, enabling ROC curves and AUC to be computed consistently with Random Forest. Here, Figure 9 summarises test-set performance from complementary perspectives. The ROC curves (Figure 9a) show the trade-off between sensitivity (true positive rate) and 1 − specificity (false positive rate) across all



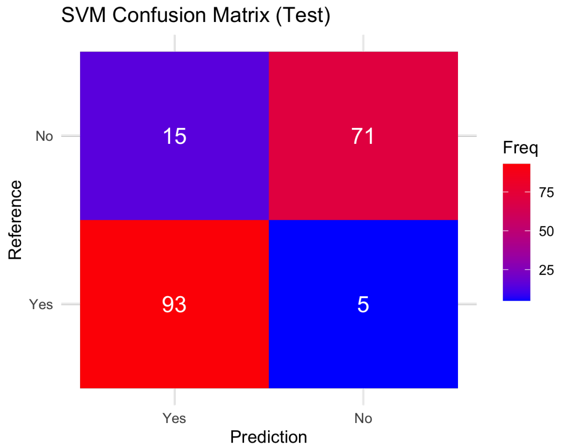
(a) RF — Train



(b) SVM — Train



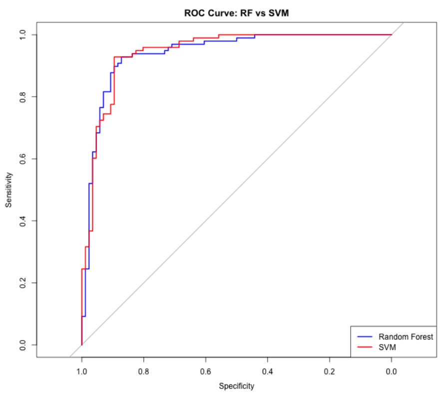
(c) RF — Test



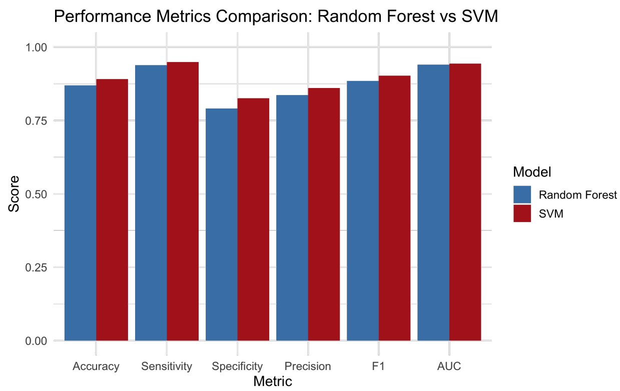
(d) SVM — Test

Figure 8: Confusion matrices on training and test sets for the selected configurations.

thresholds, where curves closer to the top-left corner and larger AUC values indicate stronger discrimination. The metric comparison (Figure 9b) reports threshold-dependent performance for the selected operating points.



(a) ROC curves on the test set.



(b) Test-set metric comparison.

Figure 9: Final test-set evaluation of Random Forest and SVM models.

Model	Accuracy	Sensitivity	Specificity	Precision	F1	AUC
RF	0.870	0.939	0.791	0.836	0.885	0.940
SVM	**0.891**	**0.949**	**0.826**	**0.861**	**0.903**	**0.944**

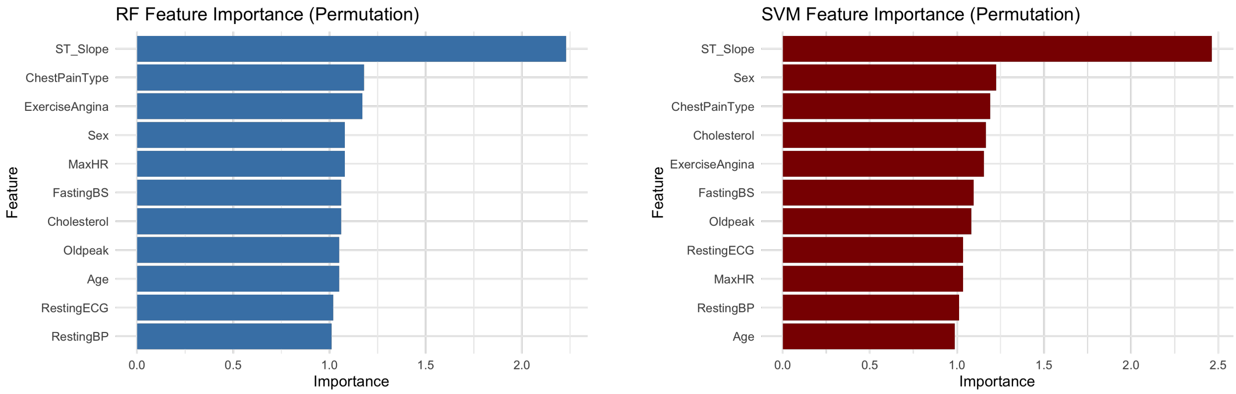
Table 4: Test-set performance metrics.

5 XAI Interpretation and Comparison

This section compares Random Forest and Support Vector Machine models using model-agnostic XAI to identify key predictors and explain their global importance and effects via permutation importance and Accumulated Local Effects (ALE) [11, 12].

5.1 Global Feature Importance (Permutation Importance)

Permutation Feature Importance (PFI) measures the decrease in model performance when a feature is randomly permuted, with larger decreases indicating greater importance. Figure 10 shows that both RF and SVM rank ST Slope as the most influential feature, followed by Chest Pain Type, Exercise Angina, Old peak, and Max HR, indicating stable and robust predictors across models.



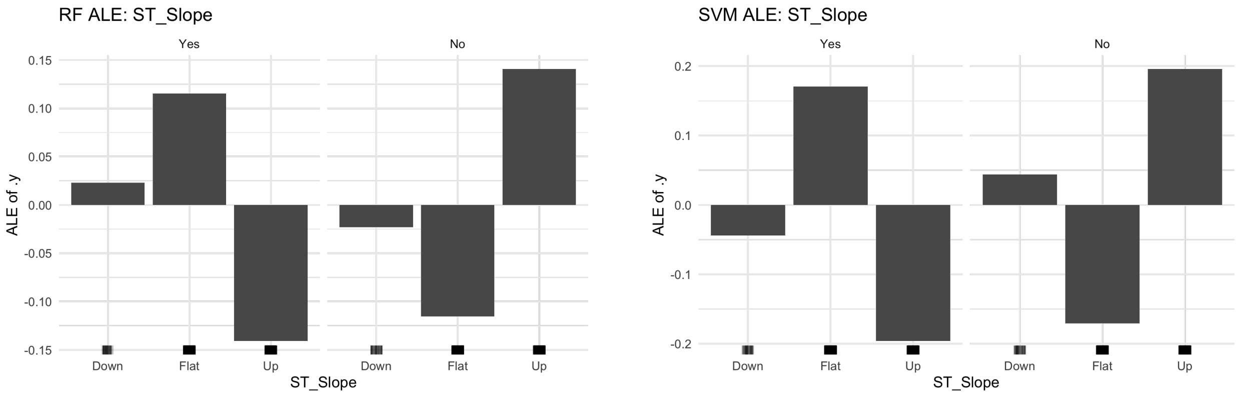
(a) Random Forest (b) Support Vector Machine

Figure 10: Permutation feature importance for RF and SVM.

5.2 Global Feature Effects (ALE)

Accumulated Local Effects (ALE) plots are used to show how important features affect predictions on average, with positive values increasing predicted risk and negative values decreasing it while accounting for feature correlations.

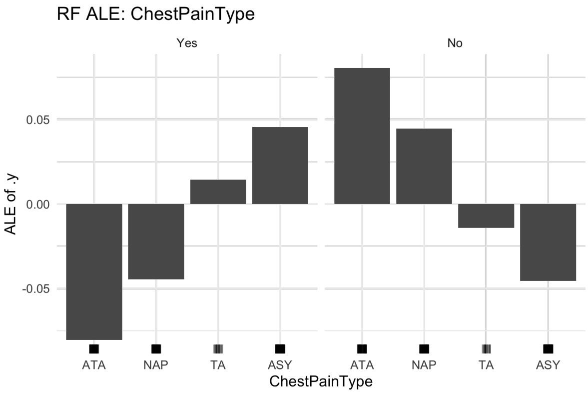
ST Slope. Figure 11 shows that an up-sloping ST segment increases predicted heart disease risk, while a down-sloping segment reduces it, consistently across both RF and SVM.



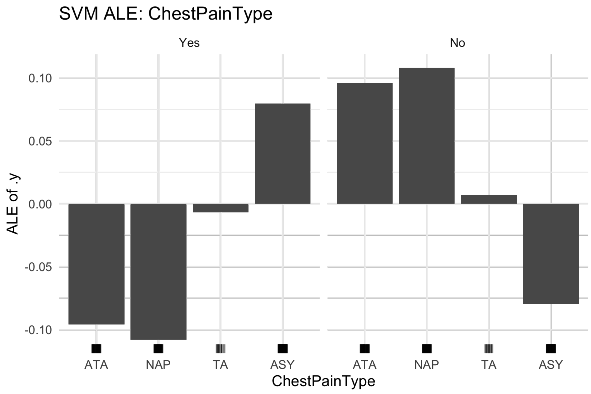
(a) RF (b) SVM

Figure 11: ALE plots for ST_Slope.

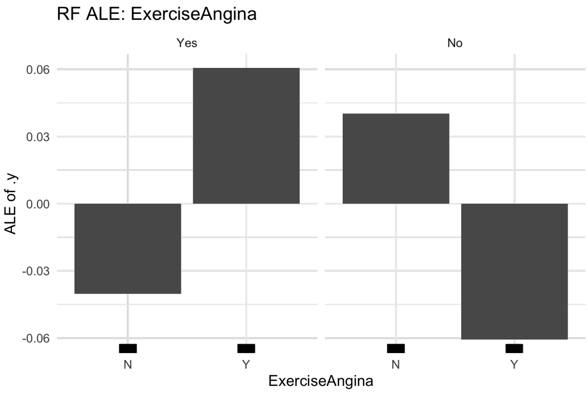
ChestPainType and ExerciseAngina. Figure 12 shows that asymptomatic chest pain and exercise-induced angina increase predicted heart disease risk, while other chest pain types and the absence of exercise angina reduce risk consistently across both models.



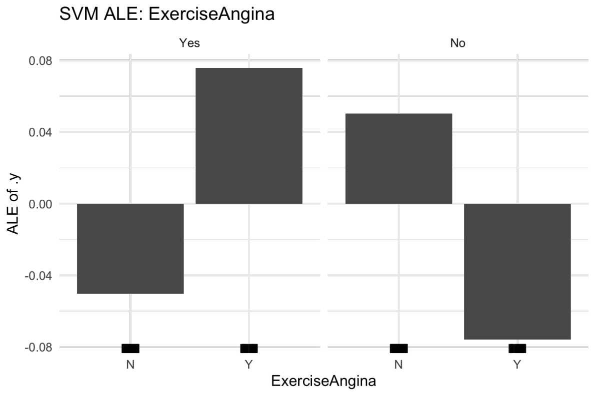
(a) RF: ChestPainType



(b) SVM: ChestPainType



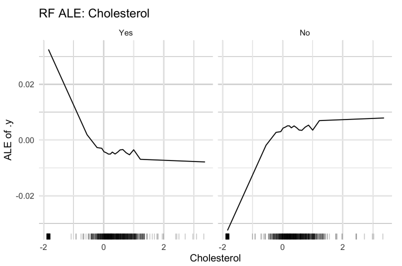
(c) RF: ExerciseAngina



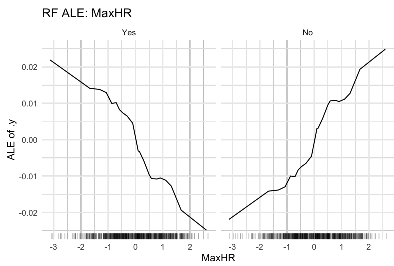
(d) SVM: ExerciseAngina

Figure 12: ALE plots for ChestPainType and ExerciseAngina.

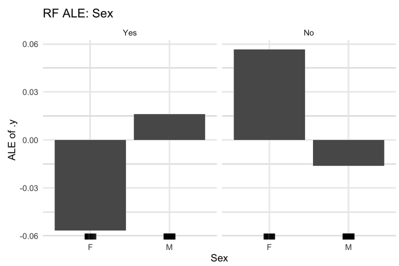
Other Features. Figure 13 shows ALE plots for the remaining predictors (**Cholesterol**, **MaxHR**, and **Sex**). These features exhibit smoother and weaker effects compared to ST_Slope and chest-pain-related variables. They support the prediction by refining risk estimates but are not primary drivers of the model decisions.



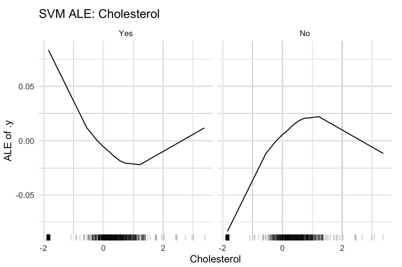
(a) RF: Cholesterol



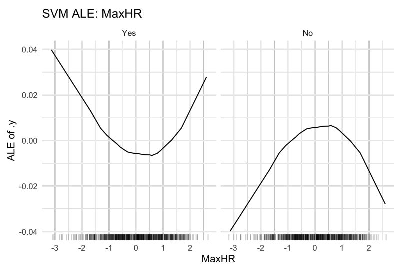
(b) RF: MaxHR



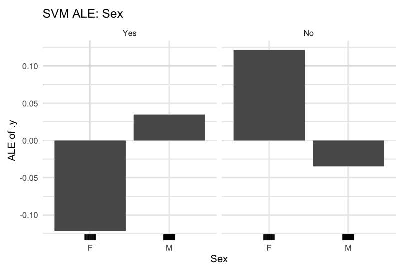
(c) RF: Sex



(d) SVM: Cholesterol



(e) SVM: MaxHR



(f) SVM: Sex

Figure 13: ALE plots for additional features arranged in a compact grid layout.

Overall Comparison. Both Random Forest and SVM demonstrate strong and consistent predictive performance across all evaluation metrics. The SVM achieves slightly higher Accuracy, Sensitivity, Specificity, Precision, F1-score, and ROC–AUC than the Random Forest, indicating a modest but consistent advantage in classification performance. Importantly, both models show comparable discrimination ability and reliable performance, supporting the robustness of the result.

Use of AI Tools

Parts of the writing, LaTeX formatting, and code refinement were supported using large language models (ChatGPT, Grok, and Gemini) [14, 15, 16]. All outputs were reviewed and validated by the authors, and the final responsibility for content, analysis, and conclusions remains with the authors.

References

[1] Kumar, R., et al. (2025). A comprehensive review of machine learning for heart disease prediction. *Frontiers in Artificial Intelligence*.

[2] Ekle, F. A., et al. (2024). Machine learning models for heart disease prediction: a comparative summary of supervised algorithms. *Journal of Healthcare Informatics Research*.

[3] Hossain, S., et al. (2024). Machine learning approach for predicting cardiovascular disease. *Cardiovascular Informatics Journal*.

[4] Rimal, Y., et al. (2025). Comparative analysis of heart disease prediction using Logistic Regression, SVM, KNN, and Random Forest. *Scientific Reports*.

[5] Baxani, R., & Edinburgh, M. (2022). Heart Disease Prediction Using Machine Learning Algorithms: Logistic Regression, Support Vector Machine and Random Forest Techniques. *International Journal of Computing and Digital Systems*.

[6] World Health Organization (2023). Cardiovascular diseases (CVDs). *WHO Fact Sheets*. `https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)`

[7] Soriano, F. (2020). *Heart Failure Prediction Dataset*. Kaggle. `https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction`

[8] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

[9] Cortes, C., Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20, 273–297.

[10] Florkowski, C. M. (2008). Sensitivity, specificity, ROC curves, and Youden's index. *Clinical Biochemistry Reviews*, 29(Suppl 1), S83–S87.

[11] Molnar, C. (2022). *Interpretable Machine Learning*. `https://christophm.github.io/interpretable-ml-book/`

[12] Apley, D. W., Zhu, J. (2019). Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models. *Journal of the Royal Statistical Society: Series B*.

[13] Lundberg, S. M., Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems (NeurIPS)*.

[14] OpenAI (2025). *ChatGPT*. Large language model. Accessed: 27 Dec 2025.

[15] xAI (2025). *Grok*. Large language model. Accessed: 27 Dec 2025.

[16] Google (2025). *Gemini*. Large language model. Accessed: 27 Dec 2025.